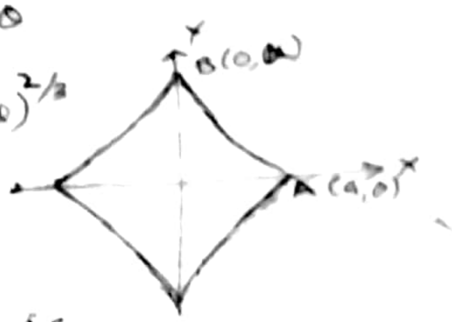
(3) Find the area of the astroid $x^{2/3}+y^{2/3}=a^{2/3}$

Put $x = a\cos^3\theta$, $y = a\sin^3\theta$

$x^{2/3}+y^{2/3} = (a\cos^3\theta)^{2/3} + (a\sin^3\theta)^{2/3}$

$= a^{2/3}(\cos^2\theta + \sin^2\theta)$

$= a^{2/3}$



$\therefore$ $x = a\cos^3\theta$, $y = a\sin^3\theta$ is the parametric form of the given curve is intersection of this curve with x-axis is given by $a\sin^3\theta = 0 \Rightarrow \sin\theta = 0$

$\Rightarrow \theta = 0, \pi$

when $\theta = 0$ $x = a$ and when $\theta = \pi$, $x = -a$

$\therefore$ Curve intersects x axis at $(a,0)$ and $(-a,0)$

III^{ly} we can prove that curve intersects y axis at $(0,a)$ and $(0,-a)$

Also curve is symmetric about both the axes

$\therefore$ entire area of the curve

= 4 times area of the curve in I quad

$= 4\int_{\theta=0}^{\theta=\pi/2} y\,dx$

$= 4\int_0^{\pi/2} y\frac{dx}{d\theta}\,d\theta$

$= 4\int_0^{\pi/2}(a\sin^3\theta)(-3a\cos^2\theta)\sin\theta\,d\theta$

$= -12a^2\int_0^{\pi/2}\sin^4\theta\cos^2\theta\,d\theta$

$= -12a^2\left[\int_0^{\pi/2}\sin^4\theta(1-\sin^2\theta)\,d\theta\right]$

$= -12a^2\left[\int_0^{\pi/2}\sin^4\theta\,d\theta - \int_0^{\pi/2}\sin^6\theta\,d\theta\right]$

$= -12a^2\left[\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2} - \frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}\right]$

$= -12a^2\,\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}\left(1-\frac{5}{6}\right)$

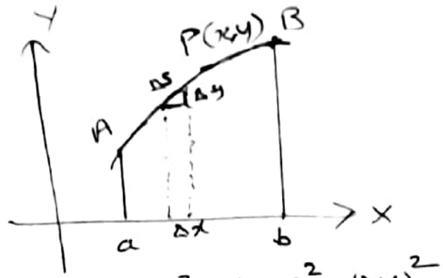
$= -12a^2\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}\times\frac{1}{6} = -\frac{3\pi a^2}{8}$

$\therefore$ Required area $= \frac{3\pi a^2}{8}$ (Taking +ve sign)

Lengths of plane curves

(i) Cartesian coordinates

If s denotes the length of the arc of the curve $y = f(x)$, measured from a fixed point A on it up to any point (x, y) on it then



$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2$

$\left(\frac{\Delta s}{\Delta x}\right)^2 = 1 + \left(\frac{\Delta y}{\Delta x}\right)^2$

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Hence the abscissa of the point A is a and that of any point B on the curve is b, then length of the arc AB is given by

$$s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

(ii) Parametric form

If the eq of the curve is given by $x = f(t)$, $y = \phi(t)$ and t_1 and t_2 are values of t corresponding to the points A and B on the curve is given by

$$s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

(iii) Polar form

If the equation of the curve is given in polar co-ordinates and θ_1 and θ_2 be the values of the vectorial angles of two points A and B on the curve, then the length of the arc AB is given by

$$s = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

Examples

(1) Find the length of the arc of the semi cubical parabola $y^2 = x^3$ extending from origin to the point $(1,1)$

The given curve is $y^2 = x^3$

Diff eq. w.r. to x

$$2y\frac{dy}{dx} = 3x^2$$

$$\therefore \frac{dy}{dx} = \frac{3x^2}{2y}$$

$\therefore$ Length of the curve from $(0,0)$ to $(1,1)$

$$= \int_0^1 \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx = \int_0^1 \sqrt{1+\frac{9x^4}{4y^2}}$$

$$= \int_0^1 \sqrt{1+\frac{9x^4}{4x^3}}\, dx = \int_0^1 \sqrt{1+\frac{9x}{4}}\, dx$$

$$= \frac{1}{2}\int_0^1 \sqrt{4+9x}\, dx = \frac{1}{2}\left[\frac{(4+9x)^{3/2}}{3/2 \cdot 9}\right]_0^1$$

$$= \frac{1}{27}\left[(4+9x)^{3/2}\right]_0^1$$

$$= \frac{1}{27}\left[13^{3/2} - 4^{3/2}\right] = \frac{1}{27}\left[13\sqrt{13} - 4\sqrt{4}\right]$$

$$= \frac{1}{27}\left[13\sqrt{13} - 8\right]$$

(2) Find the length of the curve $y = \log\sec x$ between the points given by $x = 0$ and $x = \frac{\pi}{3}$

Eq of the curve is $y = \log\sec x$

$$\frac{dy}{dx} = \frac{1}{\sec x}\cdot \sec x \cdot \tan x = \tan x$$

$\therefore$ Length of the curve from $x=0$ to $x = \frac{\pi}{3}$

$$= \int_0^{\pi/3} \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx = \int_0^{\pi/3} \sqrt{1+\tan^2 x}\, dx$$

$$= \int_0^{\pi/3} \sec x\, dx = \left[\log(\sec x + \tan x)\right]_0^{\pi/3}$$

$$= \log\left[\sec\frac{\pi}{3} + \tan\frac{\pi}{3}\right] - \log\left[\sec 0 + \tan 0\right]$$

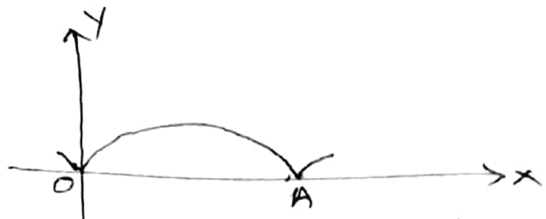
$= \log(2+\sqrt{3}) - \log(1+0)$

$= \log(2+\sqrt{3})$

(3) Show that $8a$ is the length of one arch of the cycloid whose eq is given by

$x = a(t - \sin t), \quad y = a(1-\cos t)$

An arch of the cycloid is the portion between two consecutive contacts of the curve with it's base line namely x-axis



The curve meets the x-axis when $y=0$ ie

$a(1-\cos\theta) = 0$ ie $\cos\theta = 1$ ie when $\theta = 2n\pi$

Putting $n=0$ and 1 we get two consecutive points of intersection at $\theta = 0$ and $\theta = 2\pi$

Here $\dfrac{dx}{dt} = a(1-\cos t)$, $\quad \dfrac{dy}{dt} = a\sin t$

$= 2a\sin^2\frac{t}{2}$ $\qquad\qquad = 2a\sin\frac{t}{2}\cos\frac{t}{2}$

Length of one arc $= \displaystyle\int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

$= \displaystyle\int_0^{2\pi} \sqrt{4a^2\sin^4\frac{t}{2} + 4a^2\sin^2\frac{t}{2}\cos^2\frac{t}{2}}\, dt$

$= \displaystyle\int_0^{2\pi} \sqrt{4a^2\sin^2\frac{t}{2}\left(\sin^2\frac{t}{2} + \cos^2\frac{t}{2}\right)}\, dt$

$= 2a\displaystyle\int_0^{2\pi} \sin\frac{t}{2}\, dt$

$= 2a\left(-2\cos\frac{t}{2}\right)_0^{2\pi}$

$= 2a\left[-2\cos\pi + 2\cos 0\right]$

$= 2a[-2(-1) + 2]$

$= 2a(4) = 8a$

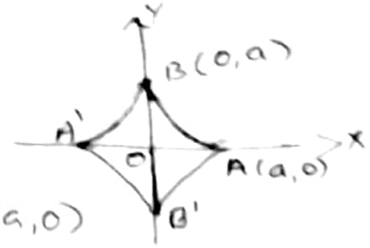
(4) Find the length of the entire curve

$x^{2/3} + y^{2/3} = a^{2/3}$

Parametric form of the given curve is

$x = a\cos^3\theta, \quad y = a\sin^3\theta$

The curve is symmetric about both the axes



It meets the x-axis at $A(a,0)$ and $A'(-a,0)$ and the y-axis at $B(0,a)$ and $B'(0,-a)$

Entire length of the curve = 4 times its length in I quadrant

First quadrant portion of the curve is bounded by the lines $\theta = 0$ and $\theta = \frac{\pi}{2}$

$\therefore$ Required length $= 4\int_{\theta=0}^{\theta=\pi/2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta$

$= 4\int_0^{\pi/2} \sqrt{(-3a\cos^2\theta\sin\theta)^2 + (3a\sin^2\theta\cos\theta)^2}\, d\theta$

$= a\int_0^{\pi/2} \sqrt{9a^2\cos^4\theta\sin^2\theta + 9a^2\sin^4\theta\cos^2\theta}\, d\theta$

$= 4\int_0^{\pi/2} \sqrt{9a^2\cos^2\theta\sin^2\theta(\cos^2\theta + \sin^2\theta)}\, d\theta$

$= 4\times 3a\int_0^{\pi/2} \sqrt{\cos^2\theta\sin^2\theta}\, d\theta$

$= 12a\int_0^{\pi/2} \sin\theta\cos\theta\, d\theta$

$= 12a\left[\frac{(\sin\theta)^2}{2}\right]_0^{\pi/2}$

$= 12a \times \frac{1}{2}\sin^2\frac{\pi}{2}$

$= 6a\times 1 = 6a$

(5) Find the perimeter of the cardioid $r = a(1-\cos\theta)$

If (r,θ) is a point on this curve $(r,-\theta)$ is also a point on it $\therefore$ It is symmetric about x-axis

∴ Perimeter of the curve = 2 times it's length abou x axis

This portion of the curve is bounded by $\theta = 0$ and $\theta = \pi$

Also $\frac{dr}{d\theta} = a \sin\theta$

$$\therefore \text{The perimeter} = 2 \times \int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$= 2 \int_0^{\pi} \sqrt{a^2(1-\cos\theta)^2 + a^2 \sin^2\theta}\, d\theta$$

$$= 2 \int_0^{\pi} \sqrt{a^2\left[1 - 2\cos\theta + \cos^2\theta + \sin^2\theta\right]}\, d\theta$$

$$= 2 \int_0^{\pi} \sqrt{2a^2(1-\cos\theta)}$$

$$= 2a \int_0^{\pi} \sqrt{2 \times 2 \sin^2\frac{\theta}{2}}\, d\theta$$

$$= 4a \int_0^{\pi} \sin\frac{\theta}{2}\, d\theta$$

$$= 4a \left(-2\cos\frac{\theta}{2}\right)_0^{\pi}$$

$$= -8a\left(\cos\frac{\pi}{2} - \cos 0\right)$$

$$= -8a(0-1) = \underline{\underline{8a}}$$

Problems

(1) Find the area bounded by $y = 5x - x^2 - 4$ and the x-axis.

(2) Find the area enclosed between the parabola $y = x^2$ and the line $2x - y + 3 = 0$

(3) Find the area of the curve $r = 3 + 2\cos\theta$.

(4) Find the length of the curve $27y^2 = 4x^3$ from $x = 0$ to $x = 3$

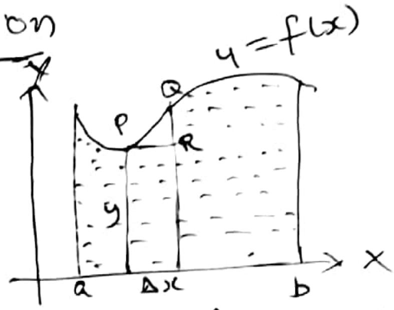
(5) Find the length of the arc of the parabola $y^2 = 4ax$ cutoff by it's latus rectum

(6) Find the length of the curve $x = t^2$, $y = t^3$ between $t = 0$ and $t = 1$

(7) Find the length of one arch of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$

(8) Find the perimeter of the cardioid $r = 2(1 - \cos\theta)$

<u>Volume of solid of revolution</u>



Consider area bounded by the curve $y = f(x)$, the x-axis and the ordinates at $x = a$ and $x = b$; when this area is revolved about x-axis A solid will be generated. We wish to find volume of this solid formed.

Consider a small strip at $P(x, y)$ of width Δx when this strip is revolved about x axis and thickness Δx a circular disc of radius y will be formed.

Volume of this circular disc $= \pi y^2 \Delta x$
$= \pi y^2\, dx$

$\therefore$ As x varies from a to b volume of the solid generated is given by

$$V = \int_a^b \pi y^2\, dx$$

<u>Note</u>: The volume of the solid formed by revolution of the area bounded by $x = g(y)$, the y-axis and the abscissa at $y = c$ and $y = d$ is given by $V = \int_c^d x^2\, dy$

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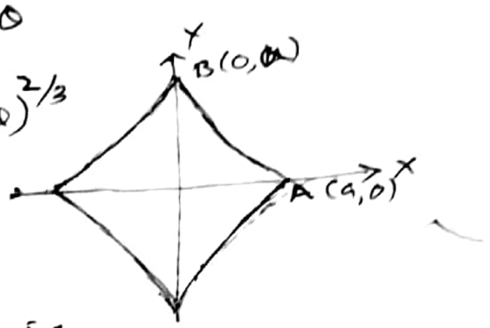

$x^{2/3}+y^{2/3} = (a\cos^3\theta)^{2/3} + (a\sin^3\theta)^{2/3}$

$= a^{2/3}(\cos^2\theta + \sin^2\theta)$

$= a^{2/3}$

$\therefore x = a\cos^3\theta,\ y = a\sin^3\theta$ is the parametric form of the given curve.

intersection of this curve with x-axis is given by $a\sin^3\theta = 0 \Rightarrow \sin\theta = 0$

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$$= -12a^2\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}\times\frac{1}{6} = -\frac{3\pi a^2}{8}$$

$\therefore$ Required area $= \dfrac{3\pi a^2}{8}$ (Taking +ve sign)

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